

Atomic Structure & the Periodic Table

Answer sheet · 36 marks total

Section A — Atoms, particles & isotopes

A1. Proton = **+1** (1); electron = **-1** (1)

A2. Atomic number = **6** (1); mass number = $6 + 6 = \mathbf{12}$ (1)

A3. Neutrons = $35 - 17 = \mathbf{18}$ (1)

A4. Atoms of the same element (same number of protons) (1) with a different number of neutrons / different mass number (1)

A5. $A_r = (79 \times 50.7 + 81 \times 49.3) \div 100$ (1) = $(4005.3 + 3993.3) \div 100 = 7998.6 \div 100$ (1) = **80.0** (1)

Section B — Electronic structure

B1. 1st shell = **2** (1); 2nd shell = **8** (1)

B2. **2, 8, 5** (2 — 1 mark if only one number is wrong)

B3. Group **7**, since it has 7 outer electrons (1); Period **3**, since it has 3 occupied shells (1)

Section C — Development of the periodic table

C1. Elements were arranged in order of atomic weight, and every 8th element had similar properties (1). Not accepted because: it forced dissimilar elements into the same group / left no gaps for undiscovered elements (any one, 1)

C2. He left gaps for elements not yet discovered (1); used the pattern to predict their properties (1); when later-discovered elements (e.g. germanium) matched these predictions, it gave scientists confidence his table reflected a real pattern (1)

C3. Atomic (proton) number (1)

C4. The same number of electrons in their outer shell (1)

Section D — Metals, non-metals & key groups

D1. Any two of: shiny; malleable/ductile; good conductors of heat & electricity; high melting/boiling points; high density (1 mark each, max 2)

D2. Any two of: dull; brittle; poor conductors of heat & electricity (insulators); low melting/boiling points (1 mark each, max 2)

D3. Noble gases have a full outer shell of electrons (1); this makes them very stable, so they do not easily gain, lose or share electrons (1)

D4. (a) sodium + water $\rightarrow$ sodium hydroxide + hydrogen (1) (b) any two of: fizzing/bubbles of gas; sodium moves/floats/melts into a ball on the surface; sodium may catch fire; the solution becomes warm (1 mark each, max 2)

D5. Going down Group 1, the outer electron is in a shell further from the nucleus (1); it is more shielded by inner shells, so it is lost more easily (1)

D6. chlorine + potassium bromide $\rightarrow$ potassium chloride + bromine (1); this is a **displacement reaction**, since chlorine is more reactive than bromine and displaces it (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any scientifically correct alternative property, observation, or reason not shown above.